

Institute of Actuaries of India

Subject ST2 – Life Insurance

March 2017 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i)**

- The appraisal value of a proprietary life insurance company is the sum of the embedded value of the company and the value to its shareholders of the future profits they expect to receive from future new business.
- The latter part of the appraisal value (value from future new business) is often referred to as the “goodwill” value of the company. **(1)**

ii)

- If a life insurance company is being valued for a merger or acquisition, the starting point will be the embedded value, as this is a measure of the profitability, which the “purchaser” or shareholders can expect to obtain from the existing in force business.
- Since the company is new and young, the embedded value may not have stabilized, and the assumptions for calculating the embedded value may also be subjective.
- Additionally, goodwill will also be subjective.
- The goodwill is more difficult to estimate, as it requires assumptions to be made on the projected sales of the company going forward.
- In particular, if the company was only incorporated two years ago, it has only made sales of two years at most. This means the projections will require significant judgement based on qualitative factors rather than past track record.
- From XYZ’s point of view projected sales will depend on the extent of similarities between the businesses of the two companies. For example if XYZ does not write with profits business the acquisition can bring a new business line that can be developed using XYZ’s greater resources.
- Also, since the ABC Life is young, its processes and practices may be in a nascent stage. Once these settle in, the profits generated may be different from the ones that are currently being observed / factored in the embedded value calculations. **(5)**

iii)**Requirement for asset share in ongoing management**

- As the company sells mainly with profits contracts, it is important that it establishes a robust framework for determining and declaring bonuses for the with profits contracts so that laid down principles can be communicated to policyholders, shareholders, regulators and other stakeholders.
- Over time, the asset share is expected to represent the amount that the insurance company has available in respect of with-profits policies.
- Although, so far, the asset shares are likely to have been negative, and hence not very relevant, since other factors such as PRE and competition may have determined the bonus rates set.
-in future asset shares are likely to be used as guidance in determining bonus rates (although there are several other considerations, e.g. PRE, competition, etc.)
- Any inaccuracy in asset shares might result in bonus rates that are unsupported by the

actual earnings from the business, leading to losses for the company.

- Or, it may result in too low bonus rates being paid out, leading to customers being treated unfairly
- ...or loss of business if competitors are able to generate higher bonuses.
- Asset shares can also be used as an effective tool for determining what might be reasonable and equitable to pay out to the different generations and classes of with-profits policyholders over time, if the calculation is granular enough.
- Therefore, inaccurate asset shares could cause customers of different tranches being treated unfairly.
- In addition the asset share is likely be the upper limit on the surrender value for all products providing a benefit on lapse (subject to the terms of the contract, and the asset share not being negative).
- When a policy is surrendered, the company makes a profit equal to the difference between the then asset share of the policy and its surrender value.
- So if inaccuracy of the asset value leads to too high a surrender value being assigned, the company could make losses.
- Particularly, if the surrender value is higher than that expected by policyholders it might encourage lapses.

Specific considerations given the acquisition

- Normal practice would be for the with profits business from ABC Life to be retained in a ring fenced sub fund in the merged company.
- This is to ensure that the merger does not adversely affect the benefit expectations and security of with-profits policyholders in ABC Life.
- It will be necessary to allocate assets to the ring-fenced sub fund.
- Asset shares will be the starting point for these calculations as well.
- And so, in order to treat policyholders fairly, the asset shares must be accurate.

(12)

iv)

The asset share is the accumulation of premiums less deductions associated with the contract, all accumulated at the actual rate of return earned on investments.

The potential deductions in the asset share include all expenditure associated with the contract, in particular:

- Expenses incurred and any commissions paid
- Cost of providing all benefits in excess of asset share – e.g. life cover or other guarantees or options granted – possibly on a smoothed, rather than current cost, basis
- Tax (if appropriate) including any reserves made for future tax liabilities
- Transfers of profit to shareholders

- Cost of any capital necessary to support contracts in the early years
- Any contribution to the free assets, which, in turn, support the smoothing of bonuses and the ability to exercise greater investment flexibility.

Additionally, the following adjustments need to be made to the asset share:

- Allocation of profits/losses on without-profits business, depending on the participation structure of the company's product funds.
- If the surrender value differs from the asset share, then the calculation could also include addition of surrender profits (or deduction of losses) from with-profits business.

(6)

[24 Marks]

Solution 2:

Investment risk

- Compared to other products, most of the investment risk from the unit-linked contracts is borne by the policyholders.
- The life insurance company will have indirect exposure if there are charges based on fund value.
- However, the investment risk for the insurance company would significantly increase if any guarantees are offered under the product, whether on maturity, surrender or death; but the risk would still be less than a non-linked contract.
- Guarantees would have to be charged which would reduce the return.
- The return to be generated would have to be significantly higher than the guarantee provided, to allow the company to deduct administrative charges and death benefit charges (if any) and still meet the guarantee.
- The insurance company may be able to manage this risk by minimizing the guarantees, although this will make the product less marketable.
- The guarantee is more likely to bite if the policyholder chooses a high return, high risk fund because such funds are inherently volatile.
- Thus different levels of guarantee may be offered for investment in different funds.
- Although administration will be very difficult in such a case, particularly if fund switching is allowed.
- If there is no investment choice offered the unit linked fund can be invested very cautiously.
- However, the returns from such assets may be lower than that expected by the policyholder leading to policyholder dissatisfaction, reputational risk, high lapses and low future sales volumes.
- The company could also consider use of derivatives, if allowed by regulations.

Regulatory changes

- Any specific regulatory restrictions on increases to charges would clearly also increase the risk.
- This type of risk will be difficult for the company to control or manage.

Expenses

- Significant risk relates to the expenses of the company being higher than the charges recovered under the unit linked contracts.
- If the unit growth is lower than expected any recovery from fund related charges towards meeting expenses may not be enough.
- Risk that the expense inflation is higher than expected may be managed by having some charges linked to an index of inflation, provided this is acceptable by the policyholders
- The charges could be structured as reviewable to reduce these expense risks, but the Company may find it difficult to review charges to fully cover expenses, especially if the unit growth has been less than expected or compared to competitor products.

Surrenders

- Surrenders occurring during the initial policy years, when the non-unit fund is negative or too small, is a significant risk under unit-linked contracts, especially when there are large initial expenses exceeding the initial charges. The same risk applies to non-linked products.
- The selective withdrawal risk is likely to be higher, due to the more open charging structures of the contracts, than under the comparable non-linked contract.
- A surrender penalty may then be imposed in order to reduce the withdrawal risk, but the surrender penalty amount may be limited by competitor designs and/or regulations

[10 marks]

Solution 3:

i)

- Smoothing is a fundamental feature of with profits business, because the benefits have a discretionary element and the company is required to be fair across different cohorts of policyholders.
- With profits products typically provide exposure to more volatile equity-type investments but the return is smoothed.
- The smoothing policy ensures that policyholders do not suffer the full detriment of rapidly falling markets, but conversely does not get full benefit from market rises.
- The average smoothed payout is usually below the average unsmoothed payout because there is a premium for smoothing that has to be paid, and this also ensures protection against overpayment of bonuses.
- To ensure all policyholders are treated fairly, the company needs to set and follow internal rules about how smoothing is applied, with the aim to minimize any subjective judgement.
- The product literature and policy illustrations should refer to smoothed returns based on this framework.

The areas that a smoothing policy would cover are as follows:

- To smooth benefits for policies maturing in a specific year, the unsmoothed terminal bonus scale would normally be expressed as a terminal bonus rate dependent on duration or year of entry with longer duration policies getting higher rates.

- A line or curve to best fit through the points representing the terminal bonus rates should be plotted.
- The smoothing policy needs to specify whether a straight line or a curve (with a given number of degrees of freedom) was allowed.
- The smoothing policy should also say whether the unsmoothed points should be weighted by the volume of business maturing at that duration.
- To smooth policy payouts for policies of the same duration maturing from one year to the next, the smoothed terminal bonus scale would be further adjusted so that the payout on policies of each duration did not change by more than x% from the payout in the previous year.
- Typically, the smoothing policy should specify the year-on-year change (x% in the point above), which might be between 5 and 20.

(7)

ii)

Overall principles

- The overall principle of terminal bonus setting would be similar for both types of with profits business.
- For bonus setting there would be two objectives to balance:
 - Bonuses should be supportable in terms of their cost (likely through asset shares)
 - Policyholders' reasonable expectations and fairness as per policy terms are to be balanced with the requirement to be competitive in the market.
- Specifically for the terminal bonus, the company needs to consider the policies terminating in the coming year due to death, surrender or maturity.
- While there is no need to consider the long-term supportability of the declared rates in respect of terminal bonuses, they should meet the average smoothing principles adopted by the company, ensuring PRE and TCF requirements.

Special considerations for accumulating with profits

- Generally, accumulated with profits business may offer less guarantees. This may mean there is more scope for discretion in setting the terminal bonus, but this depends on the specific features of the products sold by the company.
- Another important difference is the fact that on surrender, the company may have the right to apply a market value reduction ("MVR") at discretion for the accumulating with profits business (in particular for unitized accumulating with profits business). This is effectively a negative terminal bonus.

(5)

iii)

- For setting terminal bonus, the company would consider the policies due to mature in the coming period.
- These policies should be grouped according to similar characteristics, for example, policies of the same term. This will ensure equity or fairness across policyholders.

- The combined asset share for the group at the maturity date would represent an equitable payout for that group. This could be determined using the current total asset share for the policies in that group projected for the short period to the average maturity date.
- By comparing the combined asset share to the combined guaranteed benefits (original sum assured plus accumulated reversionary bonuses) an appropriate average terminal bonus rate is determined, expressed as a percentage of either sum assured or sum assured plus bonuses to date.
- This also needs to be compared to what might reasonably be expected by customers, to determine the revised bonus rates to be declared.
- The company's smoothing policy would be applied to adjust the rate if necessary.

(3)

iv) Risks arising from the economic environment

- As the market value of the investments supporting asset shares had come down, recent policy payouts including terminal bonuses may be higher than the realized value of these assets.
- The impact will depend on the smoothing carried out (asset shares allowing for smoothed terminal bonuses might be lower than actual asset shares) as well as how quickly the Company can reset the terminal bonus at lower level consistent with market values of the supporting assets.
- The company may also have had an effective hedging strategy in place (if so, the impact will be lower).
- If the rates not aligned quickly to the new supportable level of terminal bonuses, the higher cost may impact level of free assets.
- Fall in the level of free assets may impact the solvency margin as well as lead to inequity between the policyholders maturing currently and the existing policyholders.

Considerations for terminal bonuses based on the economic environment

- Some consideration is needed to be determine whether the fall in the market values is expected to be a recent one-off event.
- It may be difficult to determine whether the market movement is permanent or whether it will rebound, so if the equity market fall is significant it would be better to reduce the terminal bonus rates to avoid the risk of setting unreasonable policyholder expectations that the rates are appropriate to the new market conditions.
- If the market movement is permanent and the company takes no initial action, then some policyholders are likely to have been treated unfairly.
- If the change is expected to be a short-term fluctuation and if the markets are expected to recover then the terminal bonus scales may not need significant changes.
- If the asset values are expected to remain at lower levels for significant time, say due to events such as new government policies, significant reduction to terminal bonus rates may be required.

- If payouts were already below asset share for smoothing reasons then no action on terminal bonus rates may be needed (assuming the market movements have not pushed asset shares below the level of payouts).
- However, if payouts are now above asset share, the company needs to reduce terminal bonus rates, unless they are already zero and so cannot be reduced.
- For terminal bonus rates on surrender (if these are separate), the company should try to avoid selection against the fund so this should be further considered.
- Any reduction will need clear communication to the policyholders explaining the fall in the asset values and how the terminal bonus are consistent with its stated participating fund management principles.
- In particular, policyholders are likely to also refer to past bonus rates declared or illustrated, so the company would need to justify that the same underlying principles have led to the reduction.

(10)

[25 Marks]

Solution 4:

i)

- Appropriation price is the price at which the insurer creates units, while the company cancels units at the expropriation price.
- For both prices, the basic equity principle dictates that existing unit-holders should be in the same position before and after the transaction.
- For example, the amount of money put into the fund for each new unit appropriated, i.e. created, is such that the net asset value per unit is the same after as before the appropriation. Similar considerations apply when a company is cancelling units.
- The appropriation (expropriation) price reflects the best price at which the underlying assets can be purchased (sold), taking account of relevant dealing expenses.

(2)

ii)

Requirements under the basic equity principle

- Policy documents will normally state how the company will determine the unit prices, however there might be some statutory or other regulations on pricing units of an internal fund,
- In theory, the principle of basic equity should be followed in determining the unit prices, i.e. the interests of unit-holders not involved in a unit transaction should be unaffected by that transaction.
- On this basis, a company needs to determine unit prices only when transactions are taking place.
- However, the company would normally want to value units much more frequently – typically daily – in order to inform policyholders and for internal management.
- On days when the fund is expanding the price would be calculated on an appropriation basis; on days when the fund is contracting an expropriation basis is used.

Simplifications to the process in practice

- In practice, companies are more likely in practice use a “broad equity” approach. The basis would only be changed if there is either a trend of cash flow movements or a single significant cash flow movement against the existing basis e.g. if there is a significant net cash inflow for a fund currently being priced on a bid basis.
- Many companies will use a “management box” to minimise changes to the basis. This means that the company will try to keep a consistent basis (bid or offer basis based on whether there is a net order to buy or sell for a particular day).
- Ideally, the company should set some “limits” or rules, i.e. how it measures a significant change, which would require a review of the basis.

(6)**[8 Marks]****Solution 5:****i) Assumptions required:**

- Mortality assumption during the deferment period
- Mortality assumption post vesting
- Mortality improvement / longevity assumption
- Surrender assumption during the deferment period
- proportion of policyholders opting to take the lump sum benefit on the date of vesting
- Expenses of processing the change at vesting/claim expenses
- Expenses related to investments
- Expense assumption during deferment
- Expense assumption after vesting / expenses for annuity payment
- Expense inflation
- Discount rate

(5)**ii) The discontinuance benefit should:**

- Consider policyholders’ reasonable expectations
- Treat both surrendering and continuing policyholders equitably
- Be consistent with the annuity benefit available to continuing members
- Not exceed earned asset share in aggregate over a reasonable period
- Consider the discontinuance benefit offered by competitors
- Not be subject to frequent change, unless dictated by financial conditions
- Should be consistent with the surrender value available before and after the date of vesting
- Should be capable of being documented clearly
- Avoid selection against the insurer

(4)

iii) It does not seem to satisfy the following:

- Treat both surrendering and continuing policyholders equitably. Or at least, the policyholders do not perceive this to be the case, otherwise the take up rate would not be so high.
- It appears the policyholders perceive the benefit of discontinuance benefit to be significantly higher than the annuity benefit on continuance.
- Be consistent with the annuity benefit available to continuing members. As detailed above, this does not seem to be the case.
- Not exceed earned asset share in aggregate over a reasonable period. Given that the choice of discontinuance reduces the embedded value, it appears the benefit value exceeds the value of the policy's asset share.
- Should be consistent with the surrender value available before and after the date of vesting. Since no surrender value is available after this point, and the surrender value before this point is also significantly lower than the discontinuance benefit available on the date of vesting, it does not seem to satisfy this principle.
- Avoid selection against the insurer. Since the unhealthy lives could choose to exit just before the annuity benefit starts, it opens the insurer to anti-selection risk.

(4)

iv)

Prospective method

- Under this method, the company would calculate the present value of future benefits and expenses, net of future premiums
- Cost of the discontinuance benefit should also be allowed for, C
- The formula would be:
 - P.V. of future benefits: $10\% * P * \ddot{a}_{60}$
 - P.V. of future expenses: $e * \ddot{a}_{60}$
 - $10\% * P * \ddot{a}_{60} + e * \ddot{a}_{60} - C$
 - (need to define all terms P, C, e, etc)
- $\ddot{a}_{60}$ would need to be calculated at a rate that reflects the future expected investment return
 - ... this may be difficult to do given the long-term nature of the contract.
 - ... the result is likely to be very sensitive to the assumption chosen.
- $\ddot{a}_{60}$ would need to be calculated assuming the expected mortality for the lives who are choosing to discontinue the policy
 - ... since the option to discontinue is on the date of vesting of the immediate annuity, this might be higher than the assumed mortality for continuing lives
 - ... the formula may need to be changed to allow for future mortality improvements given the long term nature of the benefit
- e are the annual expenses expected in future
 - ... including the expenses incurred in disbursing the annuity amount

... the formula may need to be modified to allow for future expected inflation in expenses

- The above calculations are on a realistic basis
- A mark may be gained for commenting on how this is likely to produce a result that is not in line with past surrender values, although it would not strictly be an answer to the question asked.

(8)

v) The factors to be considered are:

Profitability

- The proposed benefits would entail a cost to the company which should be effectively factored into the single premium charged
- Or else, some other benefits offered, e.g. the discontinuance benefit offered or the annuity benefit offered should be reduced to ensure the product remains profitable

Marketability

- The proposed changes should be attractive to the customers; therefore the product should become more marketable
- However, if the cost of the benefit is reflected through an increase in the premium or a reduction in the other benefits, the marketability of the product may be significantly impacted
- The added benefits may not be attractive to the policyholder given its cost, since this is a retirement product, where the need for the annuity outweighs the need for a death benefit.

Competition

- Does competition offer similar benefits? If so, at what cost?
- If not, is this benefit the right trend for a retirement product?

Financing requirement

- The increased benefits will require more reserves to be maintained, thus increasing the financial strain
- However, since this would only be post vesting, by which time the asset share would be considerable, it may not have a big impact

Risk Characteristics

- The surrender benefit on an annuity product will certainly increase the anti-selection risk
- Particularly as the option can be selected at any time – for example when the policyholder's health deteriorates

- However, if the surrender benefit is equal to, or lower than the death benefit to be offered, this might reduce the anti-selection risk
- Although a dead person can't spend the money on themselves
- The death benefit would counteract the longevity risk inherent in the annuity product.

Sensitivity of profits

- As discussed above, since these benefits counter balance each other, the introduction of these could reduce sensitivity of results to withdrawals and mortality
- if priced appropriately

Admin systems

- Would the admin systems be able to cope with the introduction of the new benefits

Regulatory regime

- The regulations may not allow such a structure
- Permitting surrender of a pension and allowing the money to be spent by the policyholder may mean they have no income to live on. This is a potential reputational risk

(12)

[33 Marks]
